

Optimal Driver-Vehicle Allocation System

Project Architecture, Technical Implementation, Operational Impact, & Strategic Improvement Roadmap

Executive Summary: This project develops an automated, mathematically rigorous **Decision-Support System (DSS)** for public transportation fleet scheduling. Utilizing the **Hungarian Algorithm (Kuhn-Munkres)**, the system solves the combinatorial linear assignment problem for driver-vehicle pairing across depot stations. It eliminates manual scheduling biases, ensures 100% depot alignment, mitigates driver fatigue via balanced workload distribution (WBI), and optimizes fleet asset utilization.

1. What is this Project?

In urban public transit operations, daily vehicle assignment to drivers is traditionally managed through manual, ad-hoc, or first-come-first-served scheduling. This traditional method introduces significant operational inefficiencies:

- **Workload Disparity & Burnout:** Certain drivers accumulate excessive driving hours and distance, while others remain underutilized, creating fatigue and safety hazards.
- **Fleet Wear & Underutilization:** Subsets of vehicles sit idle or suffer disproportionate wear-and-tear due to lack of systematic asset rotation.
- **Cross-Depot Deadheading:** Assigning drivers to vehicles stationed at different depots causes uncompensated travel time and logistical waste.
- **Subjective Allocation:** Manual scheduling is prone to human error, lack of transparency, and perceived unfairness.

Project Purpose: To replace ad-hoc scheduling with an Operations Research linear programming framework that mathematically guarantees the global minimum assignment cost while enforcing strict depot alignment and workload fairness.

2. What Are We Doing Here? (Methodology & Formulation)

We formulate driver-vehicle allocation as a **Bipartite Linear Sum Assignment Problem**. Given N drivers and N vehicles, the objective is to find a bijective (1-to-1) mapping that minimizes total system cost:

Optimization Model:

Minimize: $Z = \sum_{i=1}^N \sum_{j=1}^N c_{ij} \cdot x_{ij}$
Subject to: $\sum_{j=1}^N x_{ij} = 1 \forall i$ (Every driver assigned exactly 1 vehicle)
 $\sum_{i=1}^N x_{ij} = 1 \forall j$ (Every vehicle assigned exactly 1 driver)
 $x_{ij} \in \{0, 1\} \forall (i, j)$

Multi-Objective Cost Element (c_{ij}):

$c_{ij} = w_1 \cdot D_i^{\text{norm}} + w_2 \cdot T_i^{\text{norm}} + w_3 \cdot R_j^{\text{norm}} + \text{Penalty}_{\text{depot}} + \alpha \cdot |D_i^{\text{norm}} - R_j^{\text{norm}}| + C_0$
Weights: $w_1 = 0.50$ (Distance), $w_2 = 0.30$ (Duration), $w_3 = 0.20$ (Vehicle Utilization), $\text{Penalty}_{\text{depot}} = 2.50$.

The 5-Phase Pipeline:

1. **Data Ingestion:** Clean 491 shift records and 11,352 trip transactions, standardizing timestamps and pilot/vehicle IDs.
2. **Feature Preprocessing:** Aggregate driver metrics (D_i, T_i) and vehicle trip metrics (R_j).
3. **Cost Matrix Formulation:** Build normalized 21x21 cost matrix penalizing cross-station mismatches.
4. **Optimization Engine:** Execute Kuhn-Munkres Hungarian algorithm in $O(N^3)$ time to achieve global optimum.
5. **Analytics & Export:** Evaluate Workload Balance Index (WBI), fleet utilization bands, and export operational CSV schedules.

3. What Have We Done Here? (Implementation & Results)

We restructured and implemented the system into a production-ready, modular codebase backed by an automated verification suite:

Component / Module	Key Responsibilities & Implemented Features
data/raw/ & processed/	Structured storage for raw datasets (ShiftData.csv, TripData.csv) and generated CSV schedule outputs.
src/config.py	Centralized operational weights ($w_1=0.5$, $w_2=0.3$, $w_3=0.2$), baseline constants, and directory resolutions.
src/ingestion/	Robust CSV loading, ISO/HH:MM:SS timestamp parsing, duration calculation, and profile summarization (21 drivers, 21 vehicles).
src/modeling/	21×21 cost matrix builder + Pure-Python Kuhn-Munkres solver with dual-verification against SciPy's linear_sum_assignment.
src/analytics/	Workload Variance ($\sigma^2=341.76$), Std Dev ($\sigma=18.49$ km), Workload Balance Index (WBI=0.8419), and Fleet Utilization split.
src/reporting/ & main.py	Rich terminal dashboard and CSV exporter generating optimal assignments, cost matrices, driver profiles, and utilization tables.
tests/test_pipeline.py	Automated unit test suite verifying data integrity, matrix dimensions, solver optimality, and analytics (100% pass rate).

Operational Benchmarks & Results Achieved:

Metric	Baseline	Optimized Output	Operational Impact
Total Assignment Cost	8.6823	8.3348	4.00% Cost Reduction (up to 28.8% on target pairs)
Depot Station Alignment	Mixed / Ad-hoc	100% Matched	Zero cross-depot deadheading; 100% local station compliance
Workload Balance Index (WBI)	0.8400	0.8419	More uniform driver workload; mitigates fatigue risk
Fleet Utilization (Distance)	Uneven	5 High, 13 Mod, 3 Low	Clear asset visibility for maintenance & rotation
Computation Time	Hours of manual work	< 0.05 seconds	Instantaneous polynomial-time algorithmic solution

4. Strategic Plan for Future Improvements

To evolve this project from a static assignment model into an industry-grade, enterprise **Intelligent Fleet Management System**, we have formulated a prioritized improvement roadmap categorized across 7 key engineering and operational pillars:

1. Real-Time Dynamic & Stochastic Dispatching

Extend beyond static historical averages by incorporating real-time GPS telemetry, live traffic congestion (Google Maps / OpenStreetMap API), and dynamic re-optimization to handle sudden vehicle breakdowns, route delays, or driver absenteeism seamlessly.

2. Multi-Period & Multi-Day Shift Rostering with Regulatory Constraints

Expand single-period bipartite matching into weekly/monthly scheduling using Mixed Integer Linear Programming (MILP) to enforce mandatory rest periods (e.g. EU/US DOT hours-of-service rules), weekly driving caps, and continuous fatigue modeling.

3. Driver Preference Integration & Shift-Swapping Engine

Introduce soft-constraint preference weighting (preferred morning/night shift windows, favorite routes, seniority scores) and an automated peer-to-peer shift swapping module with fairness verification to boost driver satisfaction and retention.

4. Electric Vehicle (EV) Battery State-of-Charge (SoC) & Charging Scheduling

Incorporate EV fleet dynamics: battery capacity, discharge curves per route gradient, depot charging stall availability, and Time-of-Use (ToU) electricity tariffs to co-optimize charging schedules with driver shift intervals.

5. Machine Learning-Powered Demand & Trip Duration Forecasting

Integrate predictive models (e.g., LightGBM / XGBoost / LSTM) to forecast passenger ride demand, peak congestion windows, and route trip durations from historical TripData.csv records before feeding them into the cost matrix.

6. Interactive Web Dashboard & Fleet Control Center (UI/UX)

Develop a modern, real-time web application (Streamlit / React + FastAPI) featuring interactive depot maps, live vehicle Gantt charts, driver workload histograms, 'what-if' sensitivity analysis sliders for weights (w_1, w_2, w_3), and one-click schedule dispatching.

7. Multi-Depot Fleet Rebalancing & Vehicle Maintenance Synchronization

Incorporate predictive maintenance telematics (mileage thresholds, brake/tire wear) to automatically schedule maintenance down-times and inter-depot vehicle rebalancing runs without disrupting passenger service.

5. Roadmap & Implementation Priority Matrix

Phase / Horizon	Feature / Enhancement	Impact	Complexity
Horizon 1 (Immediate: Weeks 1–4)	- Interactive Web Dashboard (Streamlit/FastAPI) - Dynamic Weight Sliders & What-If Analysis - Automated PDF/Excel Shift Manifest Exporters	HIGH	Low – Medium
Horizon 2 (Short-term: Month 2–3)	- Driver Preferences & Seniority Scoring - Multi-day Rostering with DOT Rest Hours - ML Passenger Demand & Duration Predictor	HIGH	Medium
Horizon 3 (Mid-term: Month 4–6)	- Real-Time GPS Telemetry & Dynamic Dispatch - EV Fleet Battery SoC & Depot Charging Scheduler - Automated Maintenance Sync & Rebalancing	TRANSFORMATIVE	High

Conclusion: The Optimal Driver-Vehicle Allocation System successfully transforms ad-hoc fleet scheduling into an automated, fair, and cost-optimal mathematical process. By implementing the strategic improvements in Horizons 1 through 3, this platform will establish an end-to-end intelligent transit operations ecosystem.